

Software Requirements Specification (SRS)

Project Name

Smart File System Automation and Management System

1. Project Overview

1.1 Purpose

The Smart File System Automation and Management System is designed to automate file organization, storage management, backup, duplicate detection, security, and reporting within a computer system. The system reduces manual effort, improves productivity, enhances data security, and provides intelligent file management features for individual users and organizations.

1.2 Objectives

- Automate file organization and management to reduce manual effort.
 - Improve file accessibility through efficient search and classification.
 - Enhance data security and reliability with backup and protection mechanisms.
 - Optimize storage utilization by detecting duplicate files and analyzing file sizes.
 - Provide smart reporting and monitoring features to improve overall file management efficiency.
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2. Scope

The system will monitor user-selected folders and automatically organize, manage, secure, and analyze files. It also provides reporting, backup, and intelligent search features while maintaining file activity logs.

3. Functional Requirements

FR-01 Automatic File Classification

The system shall automatically classify and move files into predefined folders based on file extensions.

File Type	Destination Folder
PDF	Documents/PDF
DOC/DOCX	Documents/Word
XLS/XLSX	Documents/Excel
PPT/PPTX	Documents/PowerPoint
JPG/PNG/GIF	Pictures
MP4/MKV	Videos
MP3/WAV	Audio
C	Programming/C
CPP	Programming/C++
PY	Programming/Python
JAVA	Programming/Java
HTML	Programming/HTML
CSS	Programming/CSS
JS	Programming/JavaScript

FR-02 File Naming Rule Validation

The system shall validate filenames according to administrator-defined naming conventions.

Features:

- Administrator-defined naming rules
 - Invalid filename detection
 - Suggested filename format
 - Prevent invalid file storage
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FR-03 File Sorting and Viewing

The system shall allow users to view and sort files using:

- File Name (A–Z)
 - File Name (Z–A)
 - File Type
 - File Size
 - Date Created
 - Date Modified
 - Oldest First
 - Newest First
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FR-04 Automatic Backup System

The system shall automatically create backup copies of selected folders.

Features

- Folder selection
 - Scheduled backup
 - Incremental backup
 - Backup logs
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FR-05 Duplicate File Management

The system shall detect duplicate files using

- File Name
- File Size
- File Hash

Actions

- Notify user
 - Remove duplicate after confirmation
 - Maintain duplicate logs
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FR-06 Recycle Bin Management

Deleted files shall be moved to the application's recycle bin.

Features

- Restore deleted files
 - Permanent deletion after configured duration
 - Restore history
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4. Additional Smart Features

FR-08 File Activity History

The system shall maintain a complete activity history for every file.

Activities recorded:

- File Created
- File Modified
- File Renamed
- File Moved
- File Deleted
- File Restored

- Timestamp
 - User Name
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FR-09 File Size Analyzer

The system shall analyze storage usage.

Features

- Largest files
 - Largest folders
 - File size distribution
 - Storage utilization
 - File count by category
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FR-10 File Password Protection

The system shall allow users to password-protect selected files.

Features

- Password protection
 - Password verification before opening
 - Change password
 - Remove password
 - Encryption support
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FR-11 Favorite Files Management

Users shall be able to mark frequently used files as Favorites.

Features

- Add to Favorites
- Remove from Favorites

- View Favorites
 - Quick Access
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FR-12 Duplicate File Suggestion Before Saving

Before saving a new file, the system shall compare it with existing files.

Comparison

- File Name
- File Size
- File Hash

Actions

- Display duplicate warning
 - Suggest overwrite or rename
 - Allow user confirmation
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FR-13 Smart File Search

The system shall provide advanced searching using multiple filters.

Search by

- File Name
 - Extension
 - File Type
 - File Size
 - Date Created
 - Date Modified
 - Keywords
 - Favorite Files
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FR-14 QR Code File Sharing

The system shall generate QR codes for selected files.

Features

- Generate QR Code
 - Local Network Sharing
 - Temporary Share Link
 - Download using QR Code
 - QR Expiration Time
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FR-15 Weekly Health Report

The system shall automatically generate weekly reports.

Report includes

- Files Added
- Files Deleted
- Storage Growth
- Duplicate Files
- Largest Files
- Backup Status
- Storage Summary

Export formats

- PDF
 - CSV
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5. Non-Functional Requirements

Performance

- File organization within 5 seconds.

- Search results displayed within 2 seconds.
- Weekly report generation within 30 seconds.

Security

- Password-protected files.
- User authentication.
- Role-based access control.
- Encrypted passwords.

Reliability

- Backup success rate of 99%.
- Accurate duplicate detection.
- Reliable activity logging.

Scalability

- Support at least 100,000 files.
- Support multiple monitored folders.

Usability.

- Easy navigation.
- Simple search functionality.

6. Hardware Requirements

- Processor: Intel i3 or higher
 - RAM: Minimum 4 GB
 - Storage: 100 GB free space
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7.Feature Priority and Development Effort

FR No.	Task	Priority	Estimated Effort
FR-01	Automatic File Classification	High	4 hrs
FR-02	File Naming Rule Validation	High	3 hrs
FR-03	File Sorting and Viewing	High	4 hrs
FR-04	Automatic Backup System	High	4 hrs
FR-05	Duplicate File Management	High	6 hrs
FR-06	Recycle Bin Management	Medium	4 hrs
FR-08	File Activity History	Medium	4 hrs
FR-09	File Size Analyzer	Medium	4 hrs
FR-10	File Password Protection	High	4 hrs
FR-11	Favorite Files Management	Low	2 hrs
FR-12	Duplicate File Suggestion Before Saving	Medium	3 hrs
FR-13	Smart File Search	High	4 hrs
FR-14	QR Code File Sharing	Low	4 hrs
FR-15	Weekly Health Report	Medium	3 hrs

8.Task Assignment to Development Team

- **Mansi**
 - FR-04: Automatic Backup System
 - FR-13: Smart File Search
 - FR-15: Weekly Health Report
- **Nirali**
 - FR-01: Automatic File Classification

- FR-10: File Password Protection
 - FR-08: File Activity History
 - **Mohini**
 - FR-05: Duplicate File Management
 - FR-03: File Sorting and Viewing
 - FR-12: Duplicate File Suggestion Before Saving
 - **Gayatri**
 - FR-02: File Naming Rule Validation
 - FR-06: Recycle Bin Management
 - FR-09: File Size Analyzer
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9. Software Requirements

Programming Language

- Python 3.x

Database

- CSV files and/or SQLite

Libraries

- watchdog
- os
- shutil
- pathlib
- hashlib
- tkinter
- pandas
- qrcode
- reportlab

- pyzipper (or equivalent for password-protected archives)
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10. Acceptance Criteria

The project shall be accepted when

1. Files are automatically organized.
 2. Naming rules are validated.
 3. Duplicate files are detected.
 4. Backup is completed successfully.
 5. Password protection works correctly.
 6. Favorite files are accessible.
 7. Smart search returns accurate results.
 8. QR codes are generated successfully.
 9. Weekly reports are generated in PDF and CSV.
 10. File activity history is maintained accurately.
 11. File size analyzer displays correct statistics.
 12. All planned test cases pass successfully.
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11. Future Enhancements

- AI-based file categorization
- Cloud backup integration
- Mobile application
- Voice-controlled file search
- OCR-based document search
- Email notifications
- Facial authentication
- Multi-device synchronization